

PUBLIC-PROJECT MECHANISM DESIGN • CERTIFICATE-FIRST RESEARCH

When the “fair” funding rule isn’t

An exact map of how small groups can break the textbook rule for funding shared projects — and by exactly how much.

A coalition of just 2–3 people breaks the rule in most tested cases — and pays \$0 for it.

Mohit Mendiratta • Independent research • SSRN Abstract ID 7293498

github.com/jammysunshine/research-showcase/tree/main/67-when-fair-funding-isnt

How do you fairly split the bill for something everyone benefits from?

- Picture an HOA board deciding whether to fund a new roof, a neighborhood group crowdfunding a footbridge, or a DAO voting to spend from a shared treasury.
- The standard textbook rule (the “pivotal mechanism”) says: build the project if pledges cover the cost, and charge each person only the smallest amount that was actually necessary for their vote to matter.
- This rule is provably honest for individuals — nobody benefits from lying about what they value the project at, acting alone.
- This paper asks the next question nobody had answered with an exact formula: what happens when a small group coordinates their pledges together?

Groups can game the textbook rule — this paper proves exactly when

- The “fair” critical-payment rule is not fair against coordinated groups: in the tested cases, coalitions of just two or three people broke it in the large majority of configurations.
- This paper proves — not just tests — exactly when that happens: whenever the project's cost is low enough that any one member's pledge is dispensable, the whole group can jointly game the price to zero.
- A simple, checkable formula tells a designer, before launch, whether their specific group size, pledge cap, and project cost are exposed.

99.2%

of a broad configuration sweep is
manipulable by a coalition of just two

145

independently brute-force-checked cases
match the closed-form formula exactly

Three neighbors, a \$3,000 footbridge

The rule only bills you for being the deciding vote. Rig it so you never are — even by lying — and you pay \$0.

1

HONESTLY

$\$0 + \$1,000 + \$1,000 = \$2,000 < \$3,000$ cost $\rightarrow$ nothing built,
nobody pays.

2

THEY COLLUDE

All three instead claim \$1,500 each = $\$4,500 \geq \$3,000 \rightarrow$ the
bridge gets built.

3

WHY IS EACH NEIGHBOR CHARGED \$0?

Drop any ONE neighbor's \$1,500 claim — the other two still total \$3,000, already enough on its own. So no single neighbor was ever needed, and each one's bill is $\$3,000 - \$3,000 = \$0$.

4

SO WHO ACTUALLY PAYS THE \$3,000?

Nobody among the three — that's the exploit. The rule promises to be self-funding (“no subsidy”), and here that promise silently breaks: the \$3,000 has to come from outside the vote — an HOA reserve fund, a grant, a contractor eating a loss — or, if nobody covers it, the bridge the rule says to build never actually gets built. Either way, three neighbors who together handed over \$0 walk away with credit for a real \$3,000 asset.

This is not a bug in one bad rule — it is the standard, textbook “pivotal mechanism” taught in every mechanism-design course.

Exhaustive, machine-checked proof — not a sampled experiment

Exact enumeration

Every possible allocation rule in the studied finite class is checked — all of them, not a sample — against six formal fairness and honesty properties.

A proven theorem

A closed-form rule predicts exactly which allocation rules survive, for any group size and pledge cap, confirmed against a larger untouched test domain.

Independent re-verification

A second, separately written program re-derives every result from raw data without reusing the first program's code — catching errors a single implementation could hide.

Exact formula for group manipulation

A closed-form expression for exactly how much a coalition can gain, derived by proof (convexity), not search — matches 145 brute-force-checked cases exactly.

A designer's checklist

The paper ends with a plain checklist: plug in your group size, pledge cap, and cost, and know before launch whether your setup is exposed.

Weak-budget-balance audit

The same exhaustive check exposes when the textbook rule itself runs a payment deficit — the flaw behind the footbridge example.

What this paper is, and deliberately is not

IS

- An exact, fully-enumerated finite theorem
- A reproducible, independently re-checked certificate
- A precise formula for coalition risk on this specific rule
- A practical pre-launch checklist for real designers

IS NOT

- A new universal impossibility theorem
- A claim about continuous (non-integer) values
- A repaired, collusion-proof mechanism
- Evidence about any specific deployed platform

Real systems already use this exact rule

- Participatory-budgeting platforms deciding which community proposal to fund.
- HOA and condo boards voting on a shared capital assessment.
- Crowdfunded neighborhood infrastructure — playgrounds, footbridges, shared solar installs.
- DAO treasuries and quadratic-funding platforms that have piloted pivotal-style payment rules.
- Anywhere a small, well-acquainted group decides together whether to build something and splits the bill by “what you were pivotal for” — this is the exact rule in question, and the paper's formula tells you if your setup is exposed.

Methodology and rigor

Certificate-first discipline

Every accepted or rejected rule ships with a machine-checkable witness; nothing is asserted from sampled performance.

Independent replay

A second implementation (`public_project_independent.py`) reconstructs every table and payment from serialized JSON without importing the primary verifier — zero mismatches across 145+ rows.

SMT cross-checks

For the neural-network/ReLU audit lane, an external exact-real Z3 solver independently confirms all reported extrema — 18 strict-bound counterexample queries, all unsat.

Preregistration

The finite-lattice theorem and its confirmation domain were committed before the larger $n=3$, $m=4$ run; coalition and false-name supplements are explicitly labeled post-hoc.

Full reproducibility

Every script, config, and SHA-256 digest is published; the entire study reruns from a single command list.

READ THE FULL PAPER

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papers.ssrn.com/sol3/papers.cfm?abstract_id=7293498

FULL CODE, DATA, AND INDEPENDENT VERIFICATION SCRIPTS

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